

Sambhav Vinay

Computer Science Undergraduate

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SUMMARY

Computer Science undergraduate with experience building scalable backend systems, distributed applications, AI-powered products, and cloud-deployed services. Strong foundation in data structures, algorithms, object-oriented programming, databases, and software engineering principles. Experienced in designing and deploying end-to-end applications using Python, Java, JavaScript/TypeScript, SQL, FastAPI, Node.js, Docker, and Google Cloud Platform. Passionate about solving large-scale engineering challenges through efficient system design, distributed systems, and machine learning.

EDUCATION

RV University, Bangalore

Expected 2028

Bachelor of Technology (Honours) in Computer Science

CGPA: 9.2/10

- Technical Lead, RUDRA (Resourceful Unit for Data Research and Analytics)

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Object-Oriented Programming
- Database Management Systems
- Software Engineering
- Operating Systems
- Computer Networks
- Probability and Statistics
- Machine Learning Fundamentals

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Distributed Systems, System Design, Debugging, Testing

Frameworks & Tools: Node.js, FastAPI, Flask, Docker, Git, Jenkins, Google Cloud Run, CrewAI

Databases: PostgreSQL, MongoDB, SQL Query Optimization

AI/ML: Machine Learning, Computer Vision, YOLOv11, ResNet-50, Agentic AI Systems

EXPERIENCE

Forward Deployed Engineer Intern

Apr 2026 – Jun 2026

TheProductWorks

- Maintained and enhanced a shared Node.js npm package used across multiple internal tools and engineering workflows.
- Developed new features and updates while ensuring backward compatibility and stability for dependent applications.
- Collaborated with product and engineering teams to resolve integration issues, debug dependencies, and improve developer experience.
- Contributed to package versioning, testing, release management, and technical documentation.
- Improved maintainability and reliability of shared software components used across the organization.

Backend Engineer Intern

Mar 2026 – Apr 2026

Clinch Technologies

- Developed backend services and components for distributed systems supporting scalable application workflows.
- Implemented APIs and service communication mechanisms enabling efficient data exchange across distributed services.
- Contributed to architectural improvements focused on scalability, reliability, and system performance.
- Participated in debugging, testing, and monitoring backend services to improve operational stability.
- Collaborated with engineers to design and implement production-ready software solutions.

Mobile Application Developer

Dec 2025 – Jan 2026

Google Developer Groups TechSprint

- Developed an AI-powered lost-and-found application using computer vision and vector similarity search.
- Designed and implemented backend services using FastAPI and deployed containerized applications on Google Cloud Run.
- Built scalable REST APIs supporting image processing, retrieval, and user interaction workflows.
- Collaborated with cross-functional teams to improve application reliability, response time, and user experience.

Technical Lead
RUDRA – RV University

Oct 2025 – Present

- Led technical initiatives and software development activities for a community of 200+ students focused on AI, machine learning, and software engineering.
- Coordinated project development workflows, technical documentation, and knowledge-sharing resources for student research teams.
- Organized workshops covering backend development, machine learning, and software engineering best practices.
- Collaborated with faculty and student teams to support research projects involving large datasets and AI systems.

PROJECTS

FinancialAudit — AI-Powered Insurance Claim Analysis System

Mar 2026 – Present

- Designed and developed a multi-agent AI system for automated insurance claim assessment and validation.
- Built backend workflows for processing structured and unstructured claim data at scale.
- Implemented rule-based verification systems to improve reliability and consistency of AI-generated decisions.
- Automated report generation and analytical pipelines supporting claim investigation processes.
- Applied software engineering principles to build maintainable and extensible agent orchestration workflows.

CalAI — Computer Vision-Based Nutrition Analysis Platform

Mar 2026 – Present

- Engineered an end-to-end computer vision pipeline for food recognition, volume estimation, and calorie prediction.
- Integrated depth estimation and image processing workflows to generate accurate nutritional assessments.
- Developed modular backend services enabling scalable inference and data processing.
- Evaluated model performance through automated testing and iterative optimization of prediction accuracy.
- Built reusable software components supporting future deployment and scalability requirements.

ReClaim — AI-Powered Lost-and-Found Platform

Dec 2025 – Jan 2026

- Built a computer vision-based object retrieval system using vector embeddings and similarity search.
- Developed backend APIs for image ingestion, feature extraction, and object matching workflows.
- Optimized retrieval performance through efficient indexing and similarity scoring techniques.
- Deployed application components using Docker and cloud-native infrastructure.

LEADERSHIP & ACTIVITIES

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- Technical Lead, RUDRA – Led technical initiatives, workshops, and collaborative software engineering projects.
 - Organized technical events, mentorship programs, and knowledge-sharing sessions for 200+ students.
 - Active contributor to AI systems, backend engineering, cloud deployment, and distributed systems projects.